

The empirical recurrence for OEIS A236732: recovering a conjecture that is not written down

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Abstract

OEIS A236732 records “Empirical recurrence of order 49”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 178$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 49, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 49 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236732 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the upper median minus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

1497, 17583, 203517, 2541745, 31912919, 411993973, 5366177421, 70784178483, ...

The entry states:

Empirical recurrence of order 49 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:30:19, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 178$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 178$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 356$ exact terms of the model returns order 49 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{49} c_i a(n-i),$$

c_1	26	c_{14}	−1833951780294	c_{27}	136849936687382938	c_{40}	−5802884236134400
c_2	4	c_{15}	3696522197250	c_{28}	−154823470104174670	c_{41}	8683325022240768
c_3	−4897	c_{16}	19347639120408	c_{29}	−199375978396153180	c_{42}	−390910646222848
c_4	23757	c_{17}	−54582774010416	c_{30}	334391635911265224	c_{43}	−926399160877056
c_5	372664	c_{18}	−138943880437603	c_{31}	153846358858481672	c_{44}	133083324710912
c_6	−2816701	c_{19}	535521570639316	c_{32}	−450652959310585080	c_{45}	44124271476736
c_7	−14378437	c_{20}	628547211611900	c_{33}	−5178580138132160	c_{46}	−9406208278528
c_8	158994487	c_{21}	−3662553078209998	c_{34}	378645845379652192	c_{47}	−582816104448
c_9	261992532	c_{22}	−1223066217903613	c_{35}	−103790841905468320	c_{48}	210390482944
c_{10}	−5397808563	c_{23}	17676996143984877	c_{36}	−191287979513077248	c_{49}	−9177137152
c_{11}	459825889	c_{24}	−4386078312870460	c_{37}	95099028756041472		
c_{12}	120334136659	c_{25}	−59688124772932962	c_{38}	52852427240566784		
c_{13}	−141698210682	c_{26}	42042260839965586	c_{39}	−40057135271457280		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 49, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $49 + 4$ terms removed returns order 49 as well, so $\deg q_{\min} = 49 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $49 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 49 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A236732.
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